The Prime Staircase: Superexponential Growth along the Diagonal of the Hierarchy of Higher-Order Primes

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Abstract. Let p_n denote the n-th prime number and let $f(n) = p_n$ be the *prime-indexing map*. Iterating the selection of prime-indexed terms generates the classical hierarchy of *higher-order primes* $p_{n,k} = f^{\,k+1}(n)$, whose first levels are the primes themselves, the *superprimes* of Dressler and Parker, and their higher-order analogues. We study the diagonal of this hierarchy, the sequence $d_k = p_{k,k} = f^{\,k+1}(k)$, obtained by taking from each level k the single term whose index coincides with the level:

3, 11, 127, 1787, 52711, 1128889, 37139213, 997525853, 36294260117, 1765037224331, ...

We call it the *prime staircase*. The sequence does not appear to be catalogued in the OEIS, although its companion sequence of indices $f^k(k)$ is known (A058009) and was studied recently by Miska, Tóth and Żmija. We prove the master formula $\mathrm{li}(\log d_k) = k + 1 + \mathrm{li}(\log k) + \mathcal{O}(\log\log k)$, whence

$$\log d_k = k(\log k + \log\log k - 1) + (\log k)(1 + \mathrm{li}(\log k)) + \mathcal{O}\left(\frac{k(\log\log k)^2}{\log k}\right).$$

In particular $\log d_k \sim k \log k \sim p_k$: the logarithm of the k-th staircase prime is asymptotic to the k-th prime itself. We also determine the computational horizon of the sequence: $d_1, \ldots, d_{10}$ are computed exactly here (the largest, $d_{10} = 1\,765\,037\,224\,331$, in under four minutes on two cores), $d_{11}, \ldots, d_{14}$ are bracketed rigorously, and the practical limits of sieve-based and prime-counting-based methods are analyzed.

Keywords: prime-indexed primes; superprimes; higher-order primes; order of primeness; prime staircase; asymptotic growth.

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1 Introduction

Let $p_1 = 2$, $p_2 = 3$, $p_3 = 5, \ldots$ be the sequence of primes. From it, select the terms whose indices are prime: this produces the sequence $3, 5, 11, 17, 31, 41, 59, 67, \ldots$ of *prime-indexed primes*, often called *superprimes*, studied by Dressler and Parker in 1975 [5] and catalogued as A006450 in the On-Line Encyclopedia of Integer Sequences [12]. Nothing prevents iterating the construction. Selecting from the superprimes those terms whose indices are superprimes gives $5, 11, 31, 59, 127, 179, \ldots$; one step further gives $11, 31, 127, 277, 709, \ldots$, and so on. In this way one obtains an infinite hierarchy of sequences of *higher-order primes* [11, 9], each level consisting of the primes whose index chain — the chain of iterated prime-counting ancestors — begins with sufficiently many primes.

The formal definition is as follows. Write $f(n) = p_n$ for the *prime-indexing map* (also called the *primeth map*), and let f^j denote its j-fold iterate. The hierarchy $(P_k)_{k\geq 0}$ is defined by $P_0 = (p_{n,0})_{n\geq 1}$ with $p_{n,0} = p_n$, and by the recursion

$$p_{n,k+1} = p_{p_n,k} \qquad (n \geq 1,\ k \geq 0), \tag{1}$$

that is, level $k+1$ selects from level k the terms whose indices are prime. A trivial induction on k (Proposition 2.1) gives the closed form $p_{n,k} = f^{k+1}(n)$, so the k-th level of the hierarchy is simply the $(k+1)$-st image of $\mathbb{N}$ under f. The first levels are A000040, A006450, A038580, A049090, A049203 and A049202.

This note is devoted to the *diagonal* of the hierarchy. From each level k we select the single term whose index coincides with the level:

$$d_k = p_{k,k} = f^{k+1}(k) \qquad (k \geq 1). \tag{2}$$

The resulting sequence begins

$$D = (3, 11, 127, 1787, 52711, 1128889, 37139213, 997525853, 36294260117, 1765037224331, \ldots).$$

We propose for it the name *prime staircase*: to pass from d_k to d_{k+1} one climbs exactly one level of the hierarchy while advancing the index by exactly one, a diagonal staircase across the infinite table of higher-order primes. Equivalently, d_k is the k-th term of the k-th sequence in the OEIS hierarchy table above — for instance $d_4 = 1787$ is the fourth term of A049203 — and the staircase is the first superdiagonal of the iteration matrix $\mathcal{A}(n, j) = f^j(n)$ catalogued as A354967.

The construction appears to be new as an explicit object of study. Searches of the OEIS conducted in October 2026 for the initial segments $3, 11, 127, 1787, 52711$ and $11, 127, 1787, 52711$ return no match, and we are not aware of any appearance of the sequence in the literature. The nearest neighbors are: (i) the *index staircase* $m_k = f^k(k) = 2, 5, 31, 277, 5381, 87803, \ldots$, catalogued as A058009 (R. G. Wilson v, 2000) and identified as the set $\operatorname{Diag}\mathbb{P}$ in the recent work of Miska, Tóth and Żmija [9], where $d_k = p_{m_k}$ is exactly the prime indexed by the k-th term of that sequence; (ii) the full iteration matrix A354967; and (iii) the variant *A344946*, which iterates f prime(k) times starting from k and coincides with the staircase only in its first two terms.

Our results are of three kinds.

Structure (Section 3). We show that the staircase is a transversal of the orbits of f: every d_k sits at height $k+1$ on the orbit of k, and the orbit structure forces a curious self-intersection property — d_k belongs to the *primeth recurrence* A007097 if and only if k itself does (Proposition 3.4). We also show that, in Fernandez's terminology of the *order of primeness* [7], the staircase is an *order-of-primeness ladder*: the order of primeness of d_k equals $k+1$, up to a correction visible only when k is itself prime (Proposition 3.3). A remark connects the staircase with Matula–Goebel numbers of rooted trees.

Growth (Section 4). Our main analytic result is the master formula

$$\operatorname{li}(\log d_k) = k + 1 + \operatorname{li}(\log k) + \mathcal{O}(\log\log k),$$

from which we deduce the two-term asymptotic expansion

$$\log d_k = k(\log k + \log\log k - 1) + (\log k)\big(1 + \operatorname{li}(\log k)\big) + \mathcal{O}\left(\frac{k(\log\log k)^2}{\log k}\right),$$

with the striking corollaries $\log d_k \sim k \log k$, $d_k^{1/k} \sim e^{-1} k \log k$, and $\log d_k \sim p_k$: *the natural logarithm of the k-th staircase prime is asymptotic to the k-th prime.* Numerically this last

approximation is remarkably tight already at small k: the ratio $\log d_k/p_k$ stays inside $[0.97, 1.10]$ for $3 \le k \le 10$. This answers, for the staircase, the problem raised in [9] of finding precise asymptotics for diagonal-type subsequences of the hierarchy.

Computability (Section 5). Computing d_k amounts to evaluating the m_k-th prime, with m_k as in A058009; the cost is governed by the size of $d_k \approx (k \log k/e)^k$, so the sequence sits at the edge of what is feasible. We compute $d_1, \dots, d_{10}$ exactly — through d_9 with a plain segmented sieve on two CPU cores, and $d_{10} = 1\,765\,037\,224\,331$ with the `primesieve` library [10] in under four minutes — bracket $d_{11}, \dots, d_{14}$ rigorously using Dusart's bounds [6] together with the known terms of A058009, and locate the practical horizon of exact computation near $k = 14$, where analytic prime-counting methods become unavoidable.

2 The hierarchy of higher-order primes

Throughout, p_n is the n-th prime, $f(n) = p_n$ is the primeth map, f^j its j-fold iterate ($f^0(n) = n$), and $\pi(x) = \#\{p \le x\}$ is the prime-counting function, so that $f = \pi^{-1}$ on $\mathbb{N}$ in the sense that p_n is the smallest x with $\pi(x) = n$. The recursion (1) defines the levels $P_k = (p_{n,k})_{n \ge 1}$ of the hierarchy.

Proposition 2.1. *For all $n \ge 1$ and $k \ge 0$,*

$$p_{n,k} = f^{k+1}(n).$$

Proof. Induction on k. For $k = 0$ the claim is $p_{n,0} = p_n = f(n)$. If it holds for k, then by (1) and the induction hypothesis, $p_{n,k+1} = p_{p_n,k} = f^{k+1}(p_n) = f^{k+1}(f(n)) = f^{k+2}(n)$. □

Thus the hierarchy is the sequence of iterated images $f(\mathbb{N}) \supset f^2(\mathbb{N}) \supset f^3(\mathbb{N}) \supset \cdots$, restricted to primes: as *sets*, the levels are nested, $P_{k+1} \subseteq P_k$, and P_k consists of the primes whose first k iterated prime-counting ancestors are prime,

$$P_k = \{q \in \mathbb{P} : \ q,\ \pi(q),\ \pi^2(q),\ \dots,\ \pi^k(q) \text{ all prime}\}.$$

In the terminology of Fernandez [7], if $F(q)$ denotes the *order of primeness* of q — the number of primes in the chain $q, \pi(q), \pi^2(q), \dots$, starting from q itself (the function A049076) — then

$$P_k = \{q \in \mathbb{P} : F(q) \ge k + 1\}. \tag{3}$$

Table 1 displays the first levels, together with their OEIS identifiers; the staircase of Section 3 is highlighted in bold along the diagonal.

Table 1: The hierarchy of higher-order primes. The entry in row k and column n is $p_{n,k} = f^{k+1}(n)$. Bold entries are the staircase values $d_k = p_{k,k}$ (one per level, on the diagonal).

level k	OEIS	index n						
		1	2	3	4	5	6	7
0	A000040	2	3	5	7	11	13	17
1	A006450	**3**	5	11	17	31	41	59
2	A038580	5	**11**	31	59	127	179	277
3	A049090	11	31	**127**	277	709	1063	1787
4	A049203	31	127	709	**1787**	5381	8527	15299
5	A049202	127	709	5381	15299	**52711**	87803	167449
6	—	709	5381	52711	167449	648391	**1128889**	2269733

The hierarchy has a substantial literature. The first level, the superprimes of Dressler and Parker [5], was studied asymptotically by Broughan and Barnett [3] and computationally by

Bayless, Klyve and Oliveira e Silva [2]; a fractal structure in their distribution is proposed by Batchko [1]. Fernandez [7] introduced the order of primeness F and observed that the record values of F are attained along the *primeth recurrence* $q_{j+1} = p_{q_j}$, $q_1 = 1$ (A007097). Miska, Tóth and Żmija [9] initiated the systematic study of the whole hierarchy and of its diagonal: they determined the convergence exponents of the level sets P_k and of the columns $\{p_n^{(k)} : k \geq 1\}$, proved that the counting function of their diagonal $\operatorname{Diag}\mathbb{P} = \{p_k^{(k)} : k \geq 1\}$ (our index staircase $m_k = f^k(k)$, A058009) satisfies $\#\{k : m_k \leq x\} \sim \log x / \log\log x$, and established the first-order asymptotics $\log m_k = k(\log k + \log\log k) + \mathcal{O}(k \log\log\log k)$. They pose as an open problem the determination of precise asymptotics for diagonal-type sequences [9, Problem 1]; the results of Section 4 answer this problem in a sharp form for the staircase d_k, and in an improved form for m_k itself (Remark 4.6).

3 The prime staircase

Definition 3.1. The *prime staircase* is the sequence $D = (d_k)_{k\geq 1}$ defined by

$$d_k = p_{k,k} = f^{k+1}(k),$$

the term of the k-th level of the hierarchy whose index equals the level. Its terms are called *staircase primes*.

Unrolling the iterates exhibits d_k as the end of a chain of $k+1$ prime evaluations starting at k; the penultimate element of the chain is the corresponding term of the index staircase A058009:

$$m_k = f^k(k), \qquad d_k = f(m_k) = p_{m_k}.$$

Table 2 lists the chains, the values m_k and d_k, and the number of decimal digits of d_k.

Table 2: The first steps of the staircase. Each row shows the orbit $k \to f(k) \to f^2(k) \to \cdots \to d_k$, the index staircase value $m_k = f^k(k)$ (A058009), the staircase prime $d_k = p_{m_k}$, and the number of decimal digits of d_k.

k	orbit of k under f, ending at d_k	m_k	d_k	digits
1	$1 \to 2 \to$ **3**	2	3	1
2	$2 \to 3 \to 5 \to$ **11**	5	11	2
3	$3 \to 5 \to 11 \to 31 \to$ **127**	31	127	3
4	$4 \to 7 \to 17 \to 59 \to 277 \to$ **1787**	277	1787	4
5	$5 \to 11 \to 31 \to 127 \to 709 \to 5381 \to$ **52711**	5381	52711	5
6	$6 \to 13 \to 41 \to 179 \to 1063 \to 8527 \to 87803 \to$ **1128889**	87803	1128889	7
7	$7 \to 17 \to 59 \to 277 \to \cdots \to 2269733 \to$ **37139213**	2269733	37139213	8
8	$8 \to 19 \to 67 \to 331 \to 2221 \to \cdots \to 50728129 \to$ **997525853**	50728129	997525853	9
9	$9 \to 23 \to 83 \to 431 \to 3001 \to \cdots \to$ **36294260117**	1559861749	36294260117	11
10	$10 \to 29 \to 109 \to 599 \to 4397 \to \cdots \to$ **1765037224331**	64988430769	1765037224331	13

Two structural features of the staircase are immediate from (2) and Proposition 2.1, and we record them for reference.

Proposition 3.2. *For every $k \geq 1$:* (i) $d_k = p_{m_k}$ *where $m_k = f^k(k)$ is the k-th term of A058009 (the diagonal* $\operatorname{Diag}\mathbb{P}$ *of [9]);* (ii) d_k *is the k-th term of the level-k sequence P_k (equivalently, (d_k) is the first superdiagonal of the iteration matrix $\mathcal{A}(n,j) = f^j(n)$, A354967).*

The staircase interacts with the hierarchy through Fernandez's order of primeness in a particularly clean way: it climbs the order ladder one rung per index.

Proposition 3.3. *Let F denote the order of primeness (A049076). Then*

$$F(d_k) = \begin{cases} k+1, & \text{if } k \text{ is composite,} \\ k+1+F(k), & \text{if } k \text{ is prime.} \end{cases}$$

Proof. The index chain of d_k is $\pi^j(d_k) = f^{k+1-j}(k)$ for $0 \le j \le k+1$, a chain of $k+2$ values $d_k, f^k(k), \ldots, f(k), k$ of which the first $k+1$ are primes (iterates of f), the last being k itself. If k is composite the chain of primes stops there, so $F(d_k) = k+1$. If k is prime, the chain continues down the index chain of k, contributing $F(k)$ further primes. □

For example, $F(d_{10}) = 11$ and $F(d_{11}) = 12 + F(11) = 12 + 4 = 16$. Note that $d_k \in P_j$ for every $j \le k$, consistent with (3): each staircase prime is a level-k prime that is simultaneously a term of every lower level.

The map f is strictly increasing and injective, with image $\mathbb{P}$; consequently its orbits organize $\mathbb{N}$ into a forest of disjoint rays. The orbit of n is $\mathcal{O}(n) = \{n, f(n), f^2(n), \ldots\}$; two orbits are either nested or disjoint, and the maximal orbits are rooted at the non-primes — at 1 and at the composites, the numbers outside the image of f. The primeth recurrence A007097 is the orbit of $1,\ 1, 2, 3, 5, 11, 31, 127, 709, 5381, 52711, \ldots$, and the sequences A057450–A057453 are the orbits of the composites $4, 6, 8, 9$. Every staircase prime lives on the orbit of its index: $d_k = f^{k+1}(k) \in \mathcal{O}(k)$. This forces the following self-intersection property, one of the pleasant hidden regularities of the construction.

Proposition 3.4. *For $k \ge 1$, the staircase prime d_k belongs to the primeth recurrence A007097 if and only if k itself belongs to A007097. More precisely, if k is the s-th term of A007097, then d_k is its $(s+k+1)$-th term.*

Proof. Orbits of f are equal or disjoint, and $d_k \in \mathcal{O}(k)$ always. Hence $d_k \in \mathcal{O}(1)$ if and only if $\mathcal{O}(k) = \mathcal{O}(1)$, that is, if and only if $k \in \mathcal{O}(1) =$ A007097. The position formula counts the $k+1$ iterates from position s onward. □

The first instances are visible in Table 2: the indices $k = 1, 2, 3, 5$ lie on A007097, and correspondingly $d_1 = 3 =$ A007097(3), $d_2 = 11 =$ A007097(5), $d_3 = 127 =$ A007097(7) and $d_5 = 52711 =$ A007097(10); the next crossing occurs at $k = 11$, then at $k = 31, 127, 709, \ldots$. Thus the staircase crosses the primeth river precisely at the primeth stepping stones, each crossing being $k+1$ stones downstream.

Remark 3.5 (Rooted trees)**.** Under the Matula–Goebel bijection between $\mathbb{N}$ and finite rooted trees — the single vertex has number 1, and a root with subtrees $T_1, \ldots, T_r$ has number $\prod_i p_{m(T_i)}$ [8] — the map $f(n) = p_n$ attaches a new root below the tree of number n. Consequently d_k is the Matula–Goebel number of the rooted tree obtained by hanging the tree of number k at the far end of a path of length $k+1$: a rooted path whose tail length is dictated by the label of the attached tree. This also identifies the level sets: P_k is the set of Matula–Goebel numbers of rooted trees that contain a path of length $k+1$ issuing from the root (equivalently, of height at least $k+2$).

Remark 3.6 (First look at the growth)**.** The staircase is superexponential: the ratios d_{k+1}/d_k for $k = 1, \ldots, 9$ are

$$3.7,\ 11.5,\ 14.1,\ 29.5,\ 21.4,\ 32.9,\ 26.9,\ 36.4,\ 48.6,$$

with no visible stabilization — and indeed we shall see that $d_{k+1}/d_k \to \infty$, as is also known for the index staircase [9, Theorem 7]. The number of decimal digits of d_k gains roughly two per index (Table 2).

4 Growth of the staircase

Write

$$x_0 = k, \quad x_{j+1} = f(x_j), \qquad y_j = \log x_j,$$

so that $d_k = x_{k+1}$ and $y_{k+1} = \log d_k$. All logarithms are natural. We shall use, throughout this section, the classical bounds of Rosser and Schoenfeld [13],

$$x \log x < p_x < x(\log x + \log\log x) \qquad (x \geq 6), \tag{4}$$

and their refinements by Dusart [6]. Taking logarithms in (4) gives the basic increment estimate.

Lemma 4.1. *For $x \geq 6$,*

$$\log p_x = \log x + \log\log x + \mathcal{O}\Big(\frac{\log\log x}{\log x}\Big),$$

and the error term is nonnegative and at most $\log\big(1 + \frac{\log\log x}{\log x}\big)$. Consequently, for k large enough that $x_j \geq 6$ for all $j \leq k+1$,

$$y_{j+1} - y_j = \log y_j + \varepsilon_j, \qquad 0 \leq \varepsilon_j \leq C\,\frac{\log y_j}{y_j}, \tag{5}$$

with an absolute constant C.

Proof. The upper bound of (4) gives $\log p_x < \log x + \log(\log x + \log\log x) = \log x + \log\log x + \log\big(1 + \frac{\log\log x}{\log x}\big)$, and the lower bound gives $\log p_x > \log x + \log\log x$. This proves the first statement. For the second, note that $x_{j+1} = p_{x_j}$ with $x_j \geq x_0 = k \to \infty$ (the sequence (x_j) is increasing), and that $\log\log x_j = \log y_j$. □

The master formula is most naturally expressed through the logarithmic integral $\mathrm{li}(t) = \mathrm{PV}\int_0^t \mathrm{d}s/\log s$. Two of its standard properties are used below: the asymptotic expansion $\mathrm{li}(t) = \frac{t}{\log t}\big(1 + \frac{1}{\log t} + \mathcal{O}(\frac{1}{\log^2 t})\big)$ as $t \to \infty$, and the fact that its inverse li^{-1} is increasing with $(\mathrm{li}^{-1})'(z) = \log \mathrm{li}^{-1}(z) \sim \log z$.

Lemma 4.2. *As $z \to \infty$,*

$$\mathrm{li}^{-1}(z) = z\big(\log z + \log\log z - 1\big) + \mathcal{O}\left(\frac{z(\log\log z)^2}{\log z}\right).$$

Proof. Put $L_1 = \log z$, $L_2 = \log\log z$, and let $w = z(L_1 + L_2 - 1 + \delta)$ with $\delta = \mathcal{O}(L_2^2/L_1)$. Then $\log w = L_1 + \mathcal{O}(L_2/L_1)$, so

$$\frac{w}{\log w} = z\Big(1 + \frac{L_2 - 1 + \delta}{L_1}\Big)\Big(1 + \mathcal{O}\Big(\frac{L_2}{L_1^2}\Big)\Big) = z\Big(1 + \frac{L_2 - 1 + \delta}{L_1} + \mathcal{O}\Big(\frac{L_2^2}{L_1^2}\Big)\Big),$$

and therefore, using the expansion of li,

$$\mathrm{li}(w) = \frac{w}{\log w}\Big(1 + \frac{1}{\log w} + \mathcal{O}\Big(\frac{1}{\log^2 w}\Big)\Big) = z\Big(1 + \frac{L_2 + \delta}{L_1} + \mathcal{O}\Big(\frac{L_2^2}{L_1^2}\Big)\Big).$$

With $\delta = \mathcal{O}(L_2^2/L_1)$ this yields $\mathrm{li}(w) = z(1 + \mathcal{O}(L_2^2/L_1^2)) = z + \mathcal{O}(zL_2^2/L_1^2)$. Hence $w = \mathrm{li}^{-1}(z + \mathcal{O}(zL_2^2/L_1^2))$, and since $(\mathrm{li}^{-1})'(z) \leq 2L_1$ for large z while $zL_2^2/L_1^2 \cdot L_1 = zL_2^2/L_1$, we obtain $\mathrm{li}^{-1}(z) = w + \mathcal{O}(zL_2^2/L_1) = z(L_1 + L_2 - 1) + \mathcal{O}(zL_2^2/L_1)$. □

Theorem 4.3 (Master formula)**.** *As $k \to \infty$,*

$$\mathrm{li}(\log d_k) = k + 1 + \mathrm{li}(\log k) + \mathcal{O}(\log\log k).$$

Proof. Assume k so large that $x_j \geq 6$ for $0 \leq j \leq k+1$. By (5) and Lemma 4.1 the increment $\Delta_j = y_{j+1} - y_j$ lies in $[\log y_j,\ \log y_j + C \log y_j / y_j]$, so $\Delta_j = \log y_j\,(1+\mathcal{O}(1/y_j))$ and $\Delta_j^2 = \mathcal{O}(\log^2 y_j)$. By Taylor expansion of $1/\log t$ around $t = y_j$,

$$\operatorname{li}(y_{j+1}) - \operatorname{li}(y_j) = \int_{y_j}^{y_j+\Delta_j} \frac{\mathrm{d}t}{\log t} = \frac{\Delta_j}{\log y_j} + \mathcal{O}\Big(\frac{\Delta_j^2}{y_j \log^2 y_j}\Big) = 1 + \mathcal{O}\Big(\frac{1}{y_j}\Big).$$

Summing over $0 \leq j \leq k$ gives

$$\operatorname{li}(y_{k+1}) = k + 1 + \operatorname{li}(y_0) + \mathcal{O}\Big(\sum_{j=0}^{k} \frac{1}{y_j}\Big).$$

It remains to bound $S = \sum_{j \leq k} 1/y_j$. Since $y_{j+1} - y_j \geq \log y_j$ and $y_{j+1} - y_j \leq 2 \log y_{j+1}$ for large y, for $t \in [y_j, y_{j+1}]$ we have $1/t \geq 1/y_{j+1}$ and $\log t \leq \log y_{j+1} \leq 2 \log y_j \leq 2(y_{j+1} - y_j)$, hence

$$\frac{1}{y_{j+1}} \leq \frac{2(y_{j+1} - y_j)}{y_{j+1} \log y_{j+1}} \leq 4 \int_{y_j}^{y_{j+1}} \frac{\mathrm{d}t}{t \log t} = 4(\log\log y_{j+1} - \log\log y_j).$$

Summing the telescope and using $y_0 = \log k$, $S \leq 4(\log\log y_{k+1} - \log\log\log k) + \mathcal{O}(1/\log k)$. An induction using (5) shows $y_j = \mathcal{O}(j \log j + \log k)$, in particular $y_{k+1} = \log d_k = \mathcal{O}(k \log k)$, so $\log\log y_{k+1} \leq \log\log k + \mathcal{O}(1)$ and $S = \mathcal{O}(\log\log k)$. The claim follows since $y_{k+1} = \log d_k$. □

Corollary 4.4 (Two-term expansion)**.** *As $k \to \infty$,*

$$\log d_k = k(\log k + \log\log k - 1) + (\log k)(1 + \operatorname{li}(\log k)) + \mathcal{O}\left(\frac{k(\log\log k)^2}{\log k}\right).$$

Proof. Let $z_k = k + 1 + \operatorname{li}(\log k) + \mathcal{O}(\log\log k) = k + r_k$ with $r_k = 1 + \operatorname{li}(\log k) + \mathcal{O}(\log\log k)$. Note $\operatorname{li}(\log k) = \frac{\log k}{\log\log k}(1 + \mathcal{O}(1/\log\log k))$, so $r_k = \mathcal{O}(\log k / \log\log k)$ and, in particular, $r_k = o(k)$ and $r_k = \mathcal{O}(\log k)$. By Theorem 4.3 and $(\operatorname{li}^{-1})'(z) = \mathcal{O}(\log z)$,

$$\log d_k = \operatorname{li}^{-1}(z_k) + \mathcal{O}(\log\log k \cdot \log k).$$

Lemma 4.2 gives $\operatorname{li}^{-1}(z_k) = z_k(\log z_k + \log\log z_k - 1) + \mathcal{O}(k(\log\log k)^2/\log k)$. Now $\log z_k = \log k + \mathcal{O}(r_k/k)$, $\log\log z_k = \log\log k + \mathcal{O}(\frac{\log k}{k \log\log k})$, and expanding,

$$z_k \log z_k = k \log k + r_k \log k + \mathcal{O}\Big(\frac{r_k^2}{k}\Big), \qquad z_k \log\log z_k = k \log\log k + \mathcal{O}(r_k \log\log k),$$

$$-z_k = -k + \mathcal{O}(\log k).$$

Since $r_k \log k = (\log k)(1 + \operatorname{li}(\log k)) + \mathcal{O}(\log k \log\log k)$, and the error terms $\log k \log\log k$, $\log k$, $r_k \log\log k$ and r_k^2/k are all $\mathcal{O}(k(\log\log k)^2/\log k)$, the corollary follows. □

Corollary 4.5. *As $k \to \infty$:* (i) $\log d_k \sim k \log k$; (ii) $d_k^{1/k} \sim e^{-1} k \log k$; (iii) $\log d_k \sim p_k$, *the k-th prime;* (iv) d_k *has* $(1 + o(1))\,\frac{k}{\log 10} \log(\frac{k \log k}{e})$ *decimal digits.*

Proof. Part (i) follows from Corollary 4.4 since the error and the subleading terms are all $o(k \log k)$. Part (ii) follows by exponentiating $\frac{1}{k} \log d_k = \log k + \log\log k - 1 + o(1)$. For (iii), combine (i) with Cipolla's expansion $p_k = k(\log k + \log\log k - 1 + \mathcal{O}(\frac{\log\log k}{\log k}))$ [4]: both quantities are $k(\log k + \log\log k - 1) + o(k)$, so their ratio tends to 1. Part (iv) is (ii) expressed in base 10. □

Remark 4.6. The same argument, with $k+1$ replaced by k steps, gives the companion formula for the index staircase $\operatorname{li}(\log m_k) = k + \operatorname{li}(\log k) + \mathcal{O}(\log\log k)$, hence

$$\log m_k = k(\log k + \log\log k - 1) + \mathcal{O}\left(\frac{k(\log\log k)^2}{\log k}\right),$$

improving the error term $\mathcal{O}(k\log\log\log k)$ of [9, Theorem 9(2)], and $\log m_k = \log d_k - \log\log m_k + \mathcal{O}(1)$, so that the two diagonals are exponentially close: $d_k = m_k^{1+o(1)}$ and $d_k/(m_k\log m_k) \to 1$.

Table 3 and Figures 1 and 2 document the quality of these approximations on the exact values $d_1, \dots, d_{10}$. Three observations are worth isolating. First, the master formula of Theorem 4.3 is extraordinarily accurate: the residual $\operatorname{li}(\log d_k) - k - 1 - \operatorname{li}(\log k)$, which Theorem 4.3 only constrains to $\mathcal{O}(\log\log k)$, is already below 0.2 in absolute value for all $4 \le k \le 10$ (it oscillates between -0.013 and $+0.19$). Second, the ratio $\log d_k/p_k$ of Corollary 4.5(iii) stays in the narrow band $[0.97, 1.10]$ for all $3 \le k \le 10$ — the logarithm of the k-th staircase prime tracks the k-th prime with a few percent accuracy over four orders of magnitude of d_k. Third, the number of digits of d_k follows Corollary 4.5(iv) closely: the prediction $k\log_{10}(k\log k/e)$ gives 11.7 and 13.6 against the true values 11 and 13 at $k = 9, 10$.

Table 3: Growth of the staircase: exact values d_k, their logarithms, the leading terms of Corollary 4.5, and the k-th prime p_k. The last column is the ratio of Corollary 4.5(iii).

k	d_k	$\log d_k$	$k\log k$	$k(\log k + \log\log k - 1)$	p_k	$\log d_k/p_k$
1	3	1.099	0.00	—	2	0.549
2	11	2.398	1.386	—	3	0.799
3	127	4.844	3.296	3.583	5	0.969
4	1 787	7.488	5.545	4.555	7	1.070
5	52 711	10.873	8.047	7.614	11	0.988
6	1 128 889	13.937	10.751	10.969	13	1.072
7	37 139 213	17.430	13.621	14.558	17	1.025
8	997 525 853	20.721	16.636	18.315	19	1.091
9	36 294 260 117	24.315	19.775	22.094	23	1.057
10	1 765 037 224 331	28.199	23.026	25.779	29	0.972

The expansion of Corollary 4.4 converges rather slowly at accessible values of k — the correction $(\log k)(1+\operatorname{li}(\log k))$ is about 5.6 at $k = 10$, against an error term of size $k(\log\log k)^2/\log k \approx 3.0$ — but its structure is fully confirmed by the data: subtracting the main term $k(\log k + \log\log k - 1)$ and the correction $(\log k)(1+\operatorname{li}(\log k))$ from $\log d_k$ leaves a remainder that decreases monotonically from 3.5 at $k = 4$ to 1.2 at $k = 10$.

5 Computability: how far can one climb?

By Proposition 3.2, computing d_k exactly is the problem of evaluating the m_k-th prime, where $m_k = f^k(k)$ is itself obtained by k nested evaluations of the n-th prime at arguments growing along the orbit of k. The final step dominates the cost: by Corollary 4.5, $d_k \approx e^{-1}k\log k \cdot m_k$, and a sieve must essentially enumerate, or at least count, the primes up to $d_k \approx (k\log k/e)^{k(1+o(1))}$. Since the terms of the index staircase are known through m_{14} (A058009; the last terms are due to D. Johnson and G. Resta), the only obstacle to extending the staircase beyond d_{10} is precisely this final prime evaluation.

All computations reported here were performed on two CPU cores with 4 GB of memory. The values $d_1, \dots, d_7$ follow in milliseconds from nested calls to `sympy.prime`. An odd-only segmented sieve of our own, written in `numpy` with segments of 10^9 integers, yields $d_8 = 997\,525\,853$ in 3.5 seconds and $d_9 = 36\,294\,260\,117$ in about three minutes; its cumulative counts were validated

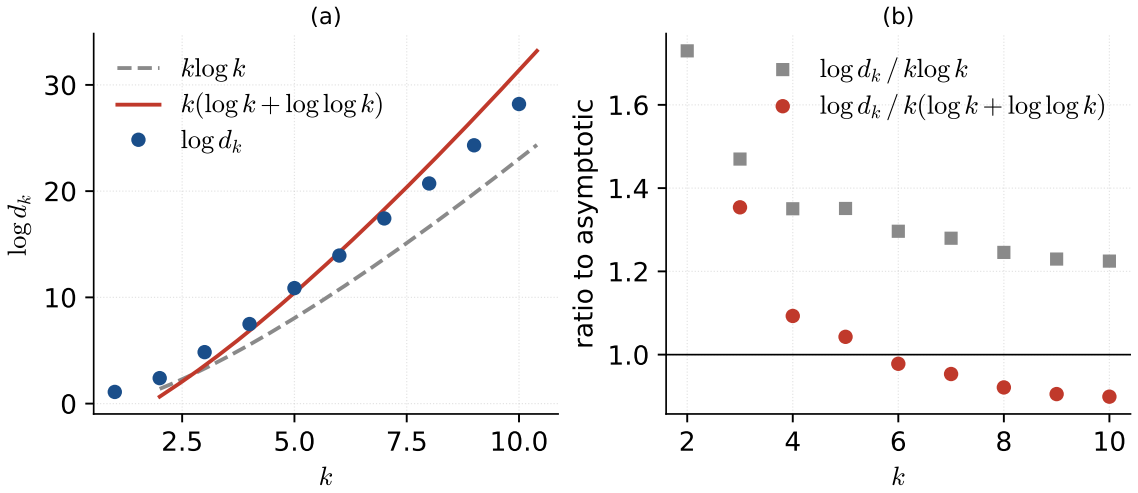


Figure 1: Growth of the staircase. (a) $\log d_k$ (dots) against the leading term $k \log k$ (dashed) and the refined leading behavior $k(\log k + \log\log k)$ (solid). (b) Ratios of $\log d_k$ to the two asymptotic expressions; the second-order correction brings the ratio close to 1, and both converge to 1 by Corollary 4.5(i).

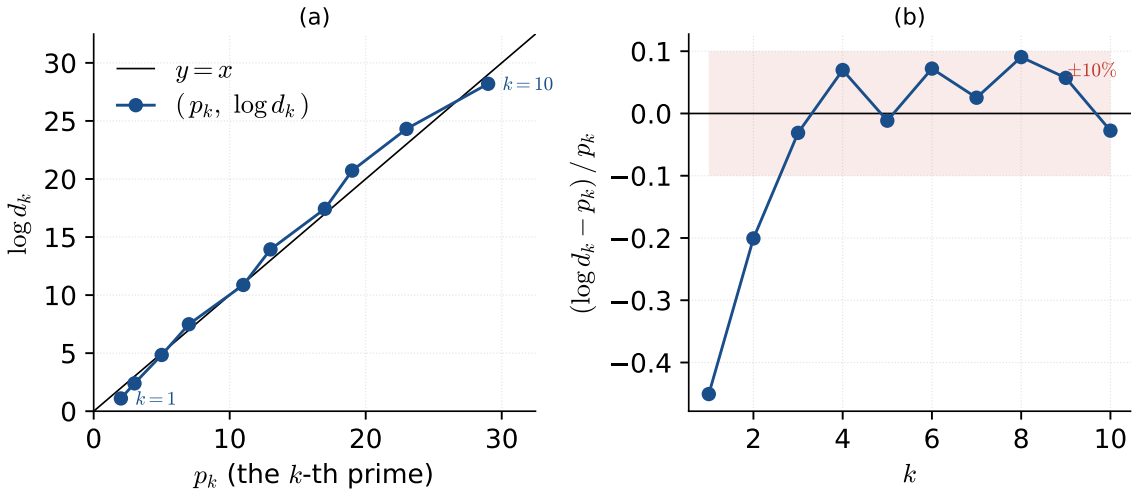


Figure 2: The approximation $\log d_k \approx p_k$ of Corollary 4.5(iii). (a) The points $(p_k, \log d_k)$ for $1 \le k \le 10$ against the line $y = x$. (b) The relative error $(\log d_k - p_k)/p_k$, which stays within $\pm 10\%$ from $k = 3$ onward.

against the classical values $\pi(10^9) = 50\,847\,534$, $\pi(10^{10}) = 455\,052\,511$ and $\pi(10^{11}) = 4\,118\,054\,813$ crossed during the run. Finally, compiling the `primesieve` library [10] from source and evaluating the 64 988 430 769-th prime gives

$$d_{10} = 1\,765\,037\,224\,331$$

in 3 minutes 43 seconds of wall time. Each value was cross-checked by at least two independent implementations.

Beyond d_{10}, exact evaluation by sieving rapidly leaves the laptop regime, but the indices are known, so rigorous brackets come almost for free from the Dusert bounds $n(\log n + \log\log n - 1) < p_n < n(\log n + \log\log n)$ for $n \geq 6$ [6]. Table 4 lists them together with the point estimates obtained by inverting li numerically at the known indices — an estimator whose accuracy is impressive: applied at m_{10} it predicts 1.765036×10^{12} against the true value 1.765037×10^{12}, a relative error below 10^{-5}.

Table 4: The upper end of the computable range. Indices m_k from A058009; the brackets are the rigorous Dusert bounds for $d_k = p_{m_k}$; the estimates invert li numerically. Digits refer to the estimates.

k	index m_k	lower bound	upper bound	estimate	digits
11	2 428 095 424 619	7.495×10^{13}	7.738×10^{13}	7.506×10^{13}	14
12	119 543 903 707 171	4.171×10^{15}	4.291×10^{15}	4.177×10^{15}	16
13	5 519 908 106 212 193	2.144×10^{17}	2.199×10^{17}	2.146×10^{17}	18
14	248 761 474 969 923 757	1.063×10^{19}	1.088×10^{19}	1.064×10^{19}	20

The cost ladder is now explicit. Sieving to x costs on the order of $x/\log x$ operations and $\mathcal{O}(\sqrt{x})$ memory, so $d_{11} \approx 7.5 \times 10^{13}$ is one to two core-hours away with a well-optimized library such as `primesieve` — an afternoon's patience rather than a research problem. The step to $d_{12} \approx 4.2 \times 10^{15}$ is two orders of magnitude: days of many-core sieving, at the limit of what is reasonable. For $d_{13} \approx 2.1 \times 10^{17}$ and $d_{14} \approx 1.1 \times 10^{19}$, sieving is hopeless, and the appropriate tool is analytic prime counting (the Meissel–Lehmer–Lagarias–Miller–Odlyzko and Deléglise–Rivat line of algorithms, as implemented in `primecount` [10]), which evaluates $\pi(x)$, and hence the n-th prime by bisection, in roughly $x^{2/3}$ operations and $\mathcal{O}(x^{1/3})$ memory; at that scale d_{13} and d_{14} become workstation computations of hours to days. Finally d_{15} would first require the index $m_{15} = p_{248\,761\,474\,969\,923\,757}$ itself, an n-th prime at $n \approx 2.5 \times 10^{17}$, and then the m_{15}-th prime with $m_{15} \approx 10^{21}$: two successive evaluations at the frontier of current record computations. We therefore place the practical horizon of exact evaluation at $k = 14$, and note the convenient rule of thumb, exact for $k \geq 11$: *the k-th staircase prime has $2k - 8$ decimal digits.*

6 Concluding remarks and open problems

The prime staircase is arguably the most economical way of walking through the hierarchy of higher-order primes: one index, one level per step. The results of this note show that despite its superexponential growth it is fully accessible to classical asymptotic analysis, and that its growth law has a memorable avatar: $\log d_k$ is asymptotic to the k-th prime itself.

We close with several problems.

Proposition 6.1. *Let $D(x) = \#\{k \geq 1 : d_k \leq x\}$. Then*

$$D(x) \sim \frac{\log x}{\log\log x} \qquad (x \to \infty).$$

Proof. By Remark 4.6, $m_k < d_k \leq m_k^{1+\varepsilon}$ for every $\varepsilon > 0$ and all large k. Hence, with $M(x) = \#\{k : m_k \leq x\}$, for large x,

$$M\big(x^{1/(1+\varepsilon)}\big) \leq D(x) \leq M(x).$$

Miska, Tóth and Żmija proved $M(x) \sim \log x / \log\log x$ [9, Theorem 12]; applying this to both bounds and letting $\varepsilon \to 0$ gives the claim. □

1. *Complete asymptotic expansion.* The proof of Theorem 4.3 strongly suggests that $\log d_k$ admits a full Cipolla-type expansion $\sum_{i\geq 0} c_i\, k\, P_i(\log k, \log\log k)$, and similarly for m_k; already the correction $(\log k)(1+\mathrm{li}(\log k))$ of Corollary 4.4 mixes the two scales $\log k$ and $\log^2 k/\log\log k$ in a way that a pure power series in $1/\log k$ cannot capture. Making this precise, and controlling the sign of the remainder, is open.

2. *Arithmetic structure.* The staircase primes show no visible congruence pattern: modulo 12 their residues are $3, 11, 7, 7, 11, 1, 5, 1, 5, 1$ for $k = 1, \ldots, 10$. Do the staircase primes equidistribute among the admissible residue classes in the (weak) sense of natural density along the sequence?

3. *Variants.* Every arithmetic self-map of $\mathbb{N}$ with image a proper subset has a hierarchy and a staircase. The composite analogue, based on iterating $n \mapsto c_n$ (the n-th composite, A280327), and the primeth-recurrence crossings of the type in Proposition 3.4 deserve study; so does the iteration-count variant A344946, which meets the staircase at $k = 1, 2$ and then diverges.

4. *Cataloguing.* The sequence

   3, 11, 127, 1787, 52711, 1128889, 37139213, 997525853, 36294260117, 1765037224331,

   with the brackets of Table 4 beyond, is proposed for inclusion in the OEIS.

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References

[1] R. G. Batchko, A prime fractal and global quasi-self-similar structure in the distribution of prime-indexed primes, arXiv:1405.2900 [math.GM], 2014.

[2] J. Bayless, D. Klyve, and T. Oliveira e Silva, New bounds and computations on prime-indexed primes, Integers **13** (2013), #A43.

[3] K. A. Broughan and A. R. Barnett, On the subsequence of primes having prime subscripts, J. Integer Seq. **12** (2009), Article 09.2.3.

[4] M. Cipolla, La determinazione asintotica dell'n-mo numero primo (in Italian), Rend. Accad. Sci. Fis. Mat. Napoli (3) **9** (1903), 132–166.

[5] R. E. Dressler and S. Parker, Primes with a prime subscript, J. ACM **22** (1975), no. 3, 380–381.

[6] P. Dusart, Estimates of some functions over primes without R.H., arXiv:1002.0442 [math.NT], 2010.

[7] N. Fernandez, An order of primeness, $F(p)$, 1999.

[8] I. Gutman and A. Ivić, On Matula numbers, Discrete Math. **150** (1996), no. 1–3, 131–142.

[9] P. Miska, J. T. Tóth, and B. Żmija, On distribution of subsequences of primes having prime indices with respect to the (R)-denseness and convergence exponent, arXiv:1908.10421 [math.NT], 2019.

[10] K. Walisch, primesieve and primecount: fast C/C++ prime number generators and counters, software, 2026.

[11] OEIS Wiki, Higher-order prime numbers, OEIS Foundation.

[12] N. J. A. Sloane (ed.), The On-Line Encyclopedia of Integer Sequences, OEIS Foundation Inc. Sequences A000040, A006450, A007097, A038580, A049090, A049076, A058009, A354967.

[13] J. B. Rosser and L. Schoenfeld, Approximate formulas for some functions of prime numbers, Illinois J. Math. **6** (1962), no. 1, 64–94.